
PT-AML: Machine learning framework to identify personalized treatments for acute myeloid leukemia

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11.12.2024

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Background

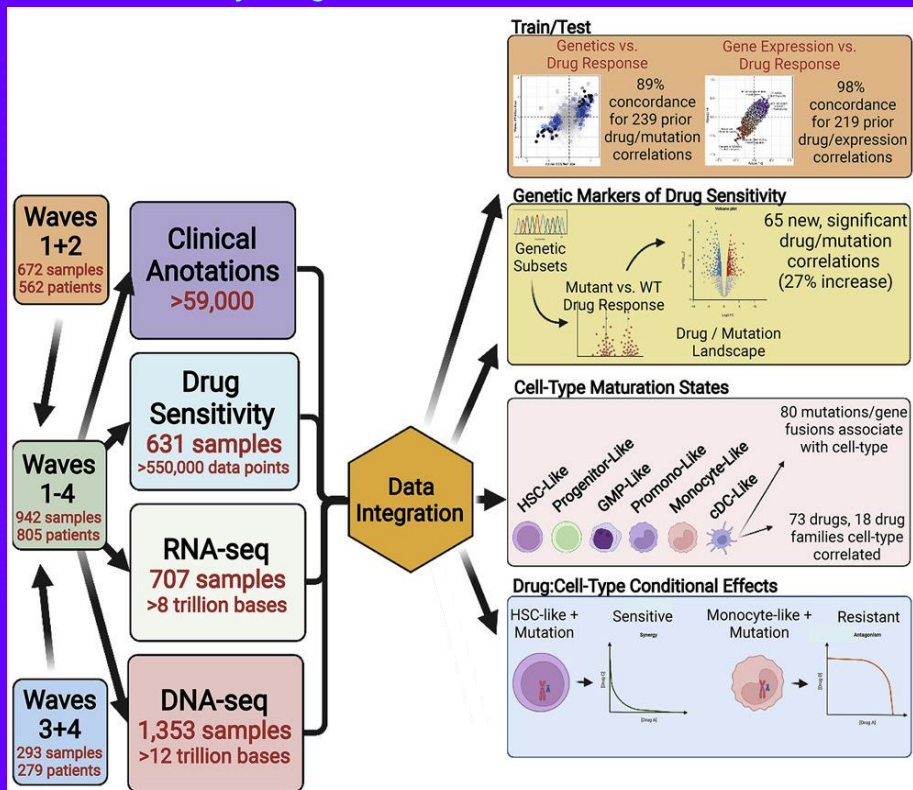
Acute Myeloid Leukemia (AML)



- Approximately 21,000 diagnoses & 10,000 AML-related deaths are reported annually in U.S.^[1]
- AML is highly heterogeneous malignancy driven by diverse genetic, celltype & molecular features^[2,3]
- Predicting drug response for individual patients is a major obstacle in AML^[4]
- The BeatAML project integrates multi-omics data with the aim to tailor therapies for AML patients^[5]
- BeatAML study analyzed 942 specimens from 805 AML patients in multiple tranches

Background

BeatAML Study Insights



Insights:

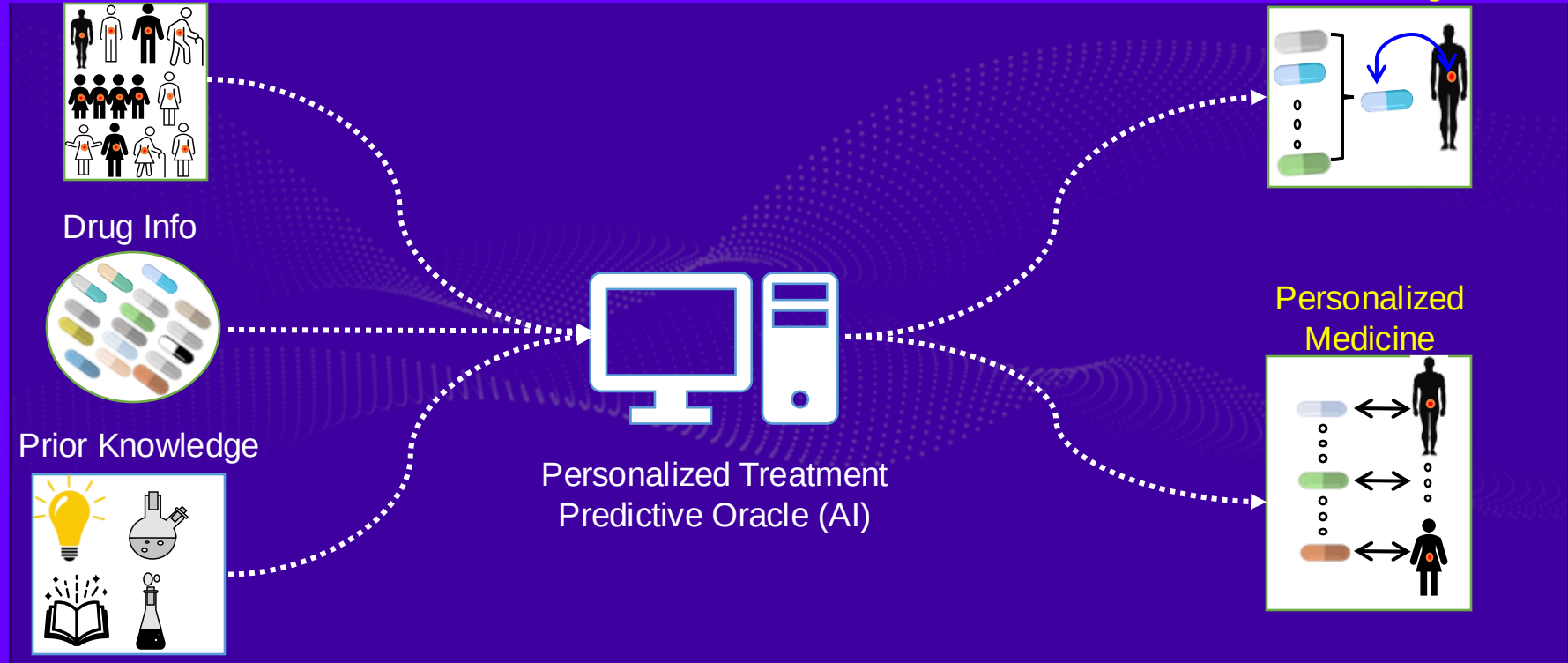
- Strong cross-cohort concordance found in mutations and gene expression analysis
- Drug response correlates with genetic mutations and AML cell differentiation state

Limitations:

- Retrospective nature of study
- **Lack of predictive modeling**

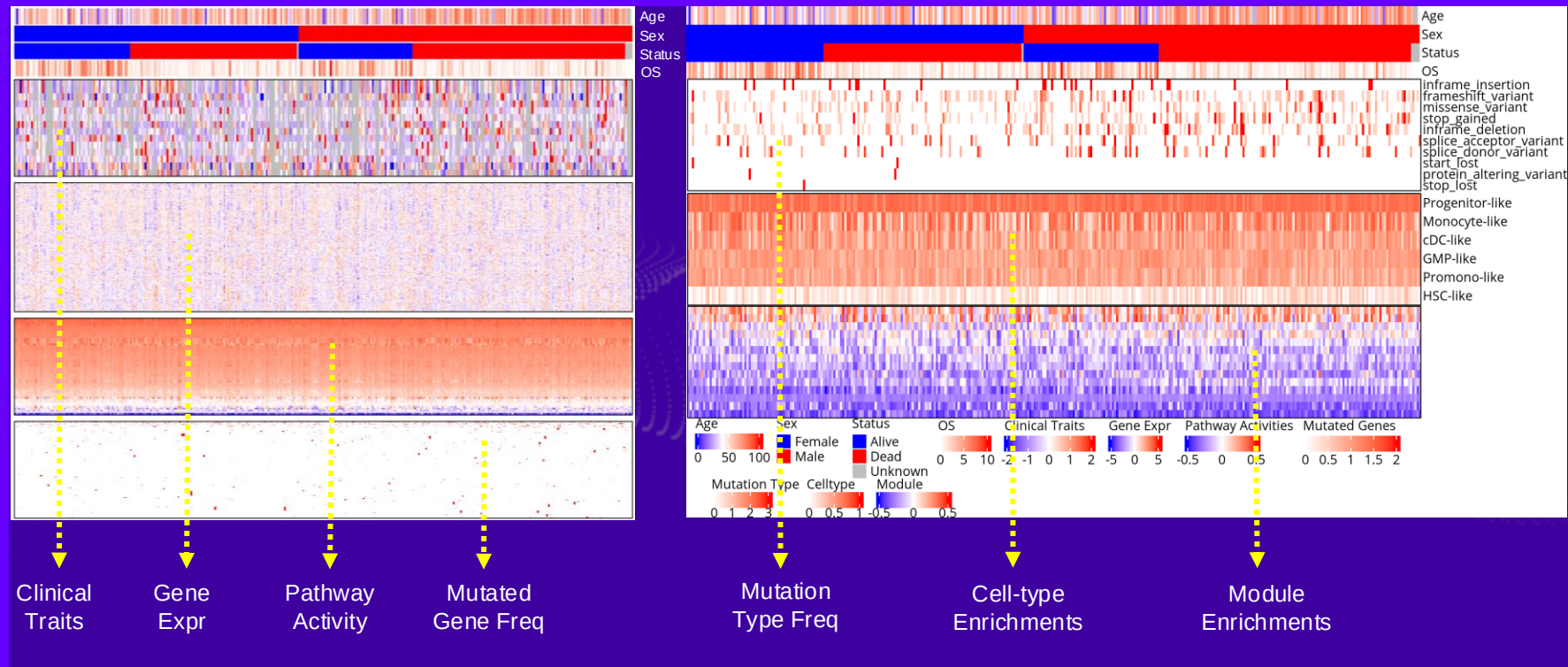
Motivations

AML Genomic Profiles



BeatAML Cohort

Multi-Omics & Clinical Annotations



- Clinical traits common across Waves & with quantitative measurements (Age, Sex, etc.) considered
- 150 most variable genes identified from RNA-Seq & kept as gene expression profile for a patient
- Frequency of mutations and frequency of mutation types were engineered features from DNA-Seq
- Different cell-state genesets^[1] were utilized with GSEA for cell-type enrichments of a given patient
- Different module genesets^[1] were utilized with GSEA for module enrichments of a given patient
- Diverse oncogenic pathway activities estimated using GSEA for a given patient
- Wave 1+2 data used as training set and Wave 3+4 data used as test set

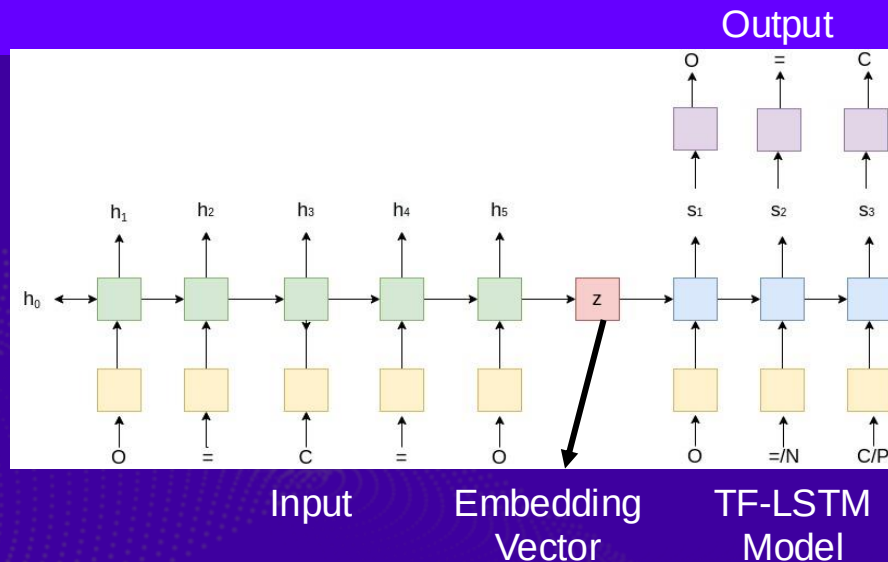
Drug Representation

Molecular Fingerprint / Embedding Representation



For drugs:

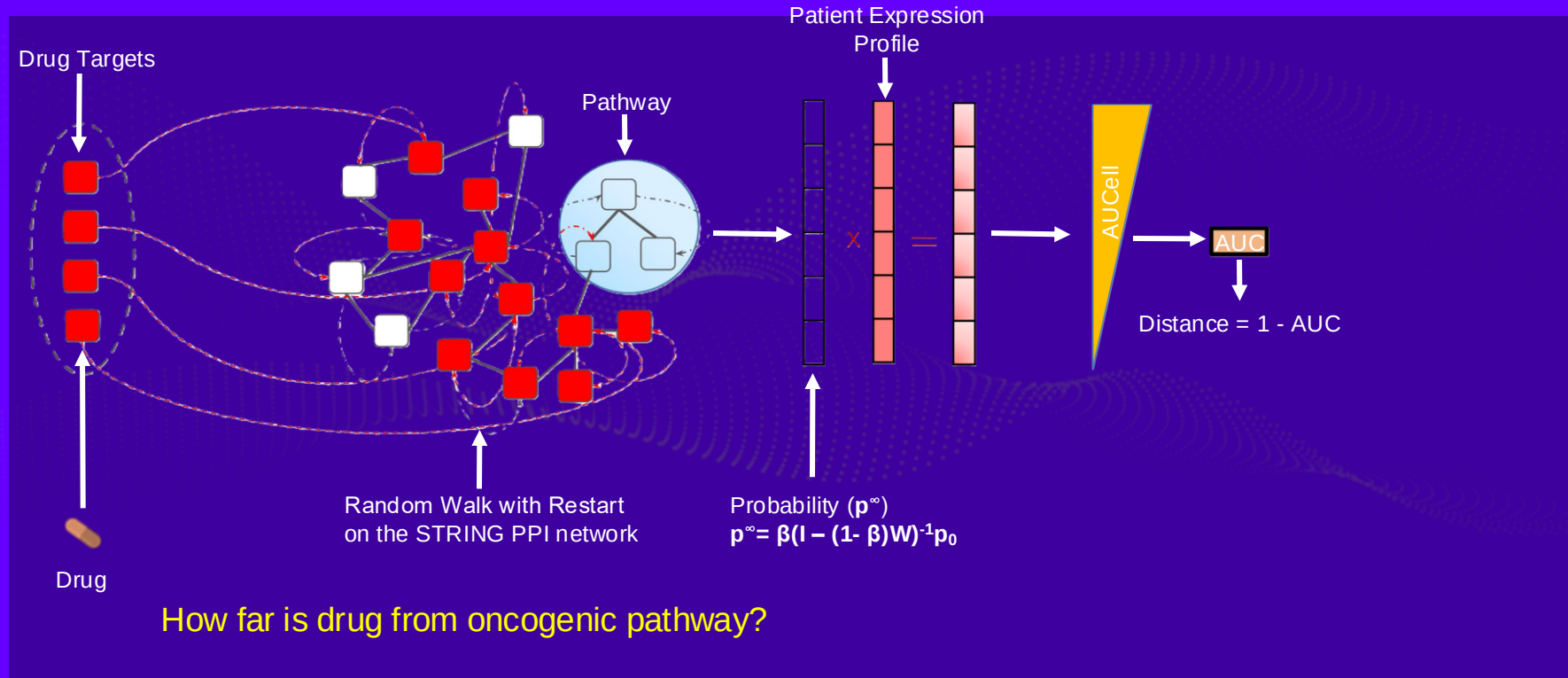
- Molecular Fingerprint (MFP) estimated using RDKIT^[1]
- SMILES-based embedding vector learnt using TF-LSTM^[2]



- Trained on 2.5 million compounds
- 96.7% reconstructed compounds are valid
- μ_{error} per sample is 0.001

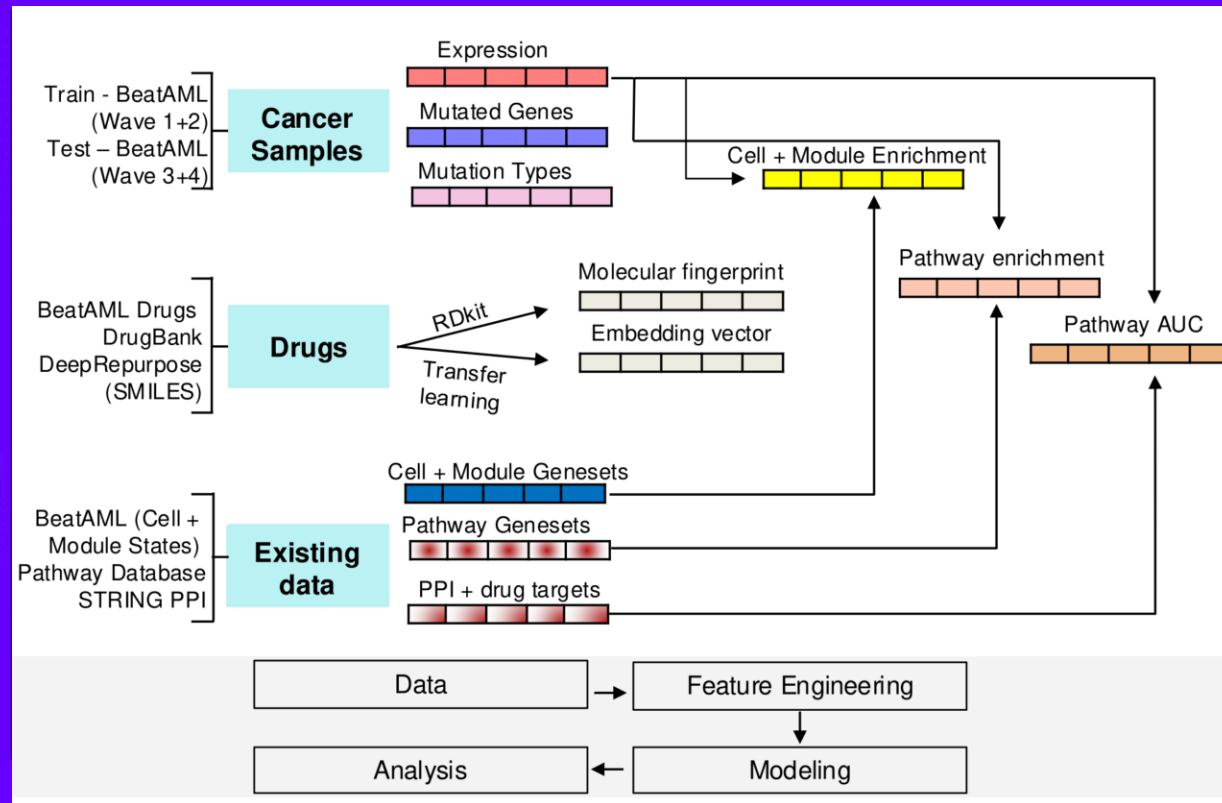
Patient-Specific Drug Oncogenic Pathway Distance

Pathway Distance (Area under curve – AUC) Estimation



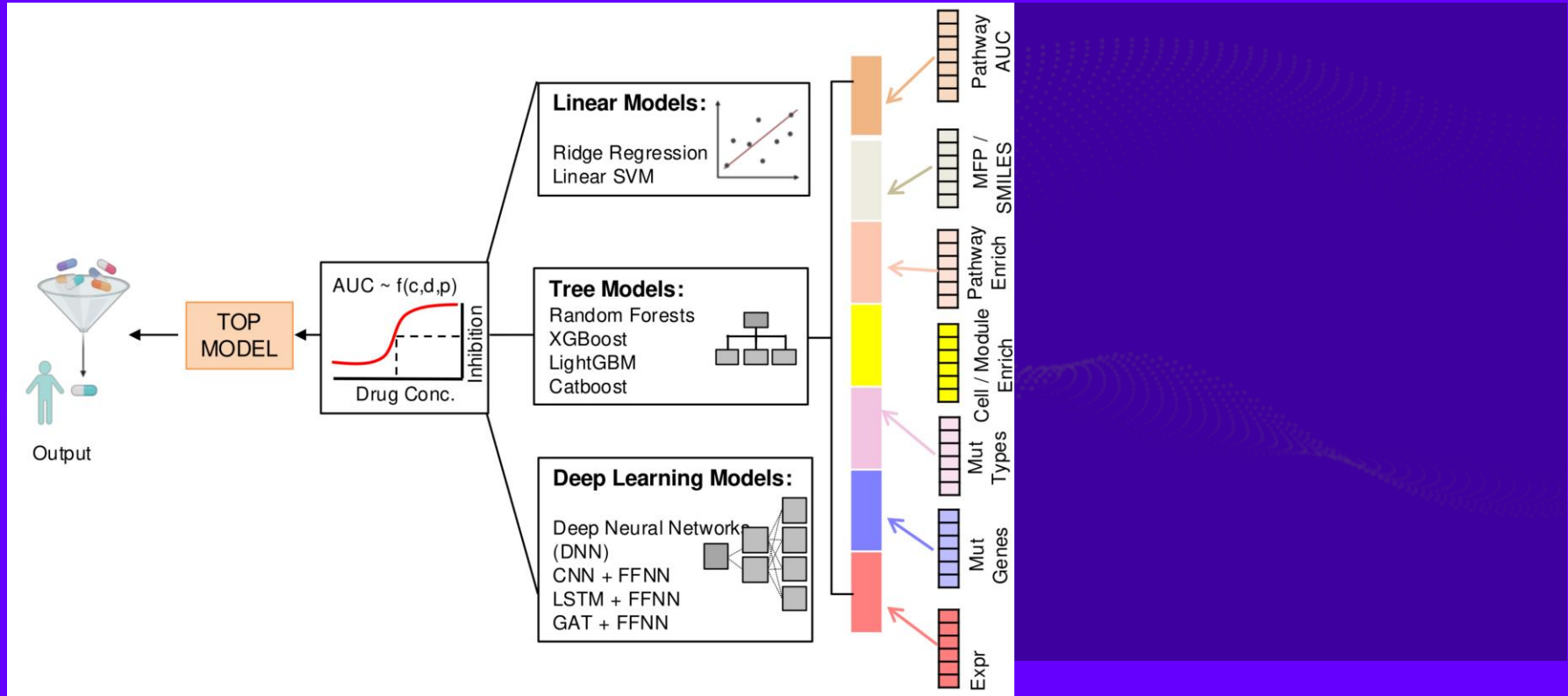
PT-AML: Personalized Treatments for AML

Feature Engineering & Workflow



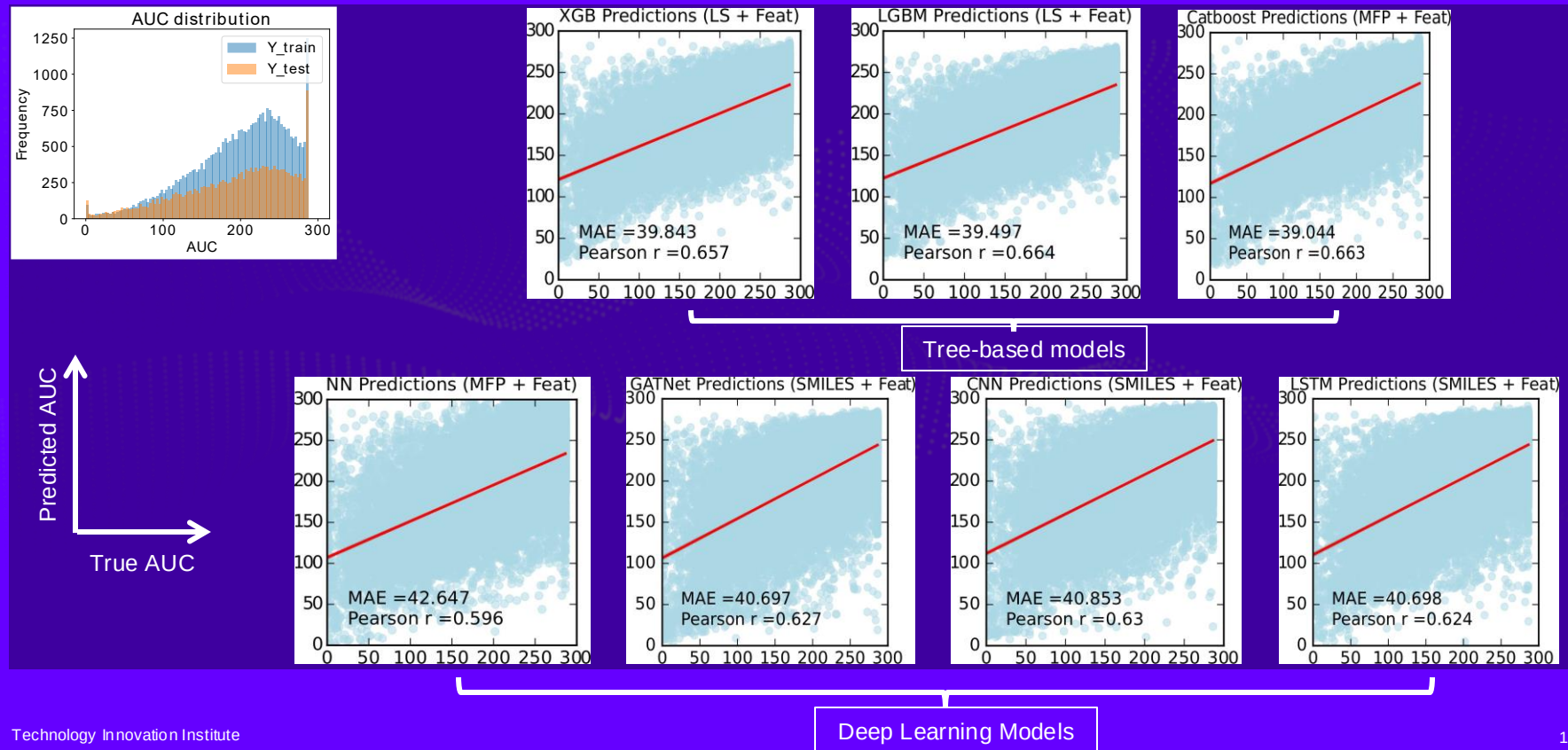
PT-AML: Personalized Treatments for AML

Model Building Workflow



PT-AML: Results

Performance on Wave 3+4 (Test Set)



PT-AML: Results

Performance on Wave 3+4 (Test Set)

Methods	Features	CV (MAE)	Test (MAE)	CV (RMSE)	Test (RMSE)	CV (R^2)	Test (R^2)	CV (r)	Test (r)	CV (sr)	Test (sr)
GLR	LS + Feat	30.22 $\pm$ 1.97	44.86	40.34 $\pm$ 2.91	60	0.56 $\pm$ 0.05	0.28	0.74 $\pm$ 0.03	0.53	0.74 $\pm$ 0.03	0.56
GLR	MFP + Feat	29.93 $\pm$ 1.92	43.48	40.10 $\pm$ 2.89	57.26	0.56 $\pm$ 0.05	0.34	0.75 $\pm$ 0.03	0.58	0.75 $\pm$ 0.03	0.58
SVR	LS + Feat	30.13 $\pm$ 1.96	44.97	40.25 $\pm$ 2.90	60.37	0.56 $\pm$ 0.05	0.28	0.75 $\pm$ 0.03	0.53	0.74 $\pm$ 0.03	0.56
SVR	MFP + Feat	29.93 $\pm$ 1.92	43.89	40.09 $\pm$ 2.89	57.73	0.56 $\pm$ 0.05	0.33	0.75 $\pm$ 0.03	0.57	0.75 $\pm$ 0.03	0.57
RF	LS + Feat	21.82 $\pm$ 0.97	39.63	29.86 $\pm$ 1.51	52.08	0.79 $\pm$ 0.02	0.44	0.89 $\pm$ 0.01	0.66	0.89 $\pm$ 0.01	0.66
RF	MFP + Feat	20.75 $\pm$ 1.22	40.16	28.29 $\pm$ 1.83	52.53	0.82 $\pm$ 0.02	0.44	0.90 $\pm$ 0.01	0.66	0.9 $\pm$ 0.01	0.65
XGBoost	LS + Feat	24.29 $\pm$ 1.21	39.84	31.94 $\pm$ 1.76	52.34	0.73 $\pm$ 0.04	0.43	0.85 $\pm$ 0.02	0.66	0.84 $\pm$ 0.02	0.65
XGBoost	MFP + Feat	22.94 $\pm$ 1.01	40.53	30.10 $\pm$ 1.45	52.88	0.78 $\pm$ 0.04	0.43	0.88 $\pm$ 0.2	0.66	0.87 $\pm$ 0.02	0.64
LightGBM	LS + Feat	21.53 $\pm$ 1.12	39.5	28.75 $\pm$ 1.63	52.09	0.79 $\pm$ 0.03	0.44	0.89 $\pm$ 0.02	0.66	0.87 $\pm$ 0.02	0.66
LightGBM	MFP + Feat	23.23 $\pm$ 1.08	40.07	31.12 $\pm$ 1.65	52.55	0.75 $\pm$ 0.04	0.43	0.86 $\pm$ 0.02	0.66	0.85 $\pm$ 0.02	0.64
Catboost	LS + Feat	26.35 $\pm$ 1.56	39.24	37.26 $\pm$ 2.51	52.45	0.62 $\pm$ 0.04	0.44	0.79 $\pm$ 0.03	0.66	0.79 $\pm$ 0.02	0.66
Catboost	MFP + Feat	21.02 $\pm$ 1.48	39.04	31.92 $\pm$ 2.43	52.03	0.73 $\pm$ 0.03	0.44	0.85 $\pm$ 0.02	0.66	0.85 $\pm$ 0.02	0.65
FFNN	LS + Feat	23.51 $\pm$ 1.43	45.15	32.09 $\pm$ 2.1	58.69	0.72 $\pm$ 0.03	0.32	0.85 $\pm$ 0.02	0.57	0.84 $\pm$ 0.02	0.55
FFNN	MFP + Feat	18.19 $\pm$ 1.05	42.65	25.42 $\pm$ 1.58	56.18	0.82 $\pm$ 0.02	0.36	0.91 $\pm$ 0.01	0.6	0.90 $\pm$ 0.01	0.59
GAT + FFNN	SMILES + Feat	26.31 $\pm$ 0.37	40.7	37.13 $\pm$ 0.64	54.9	0.64 $\pm$ 0.01	0.39	0.80 $\pm$ 0.00	0.63	0.80 $\pm$ 0.01	0.62
CNN + FFNN	SMILES + Feat	26.03 $\pm$ 0.23	40.85	37.03 $\pm$ 0.49	55.6	0.65 $\pm$ 0.01	0.4	0.80 $\pm$ 0.00	0.63	0.80 $\pm$ 0.00	0.63
LSTM + FFNN	SMILES + Feat	26.27 $\pm$ 0.42	40.7	36.87 $\pm$ 0.61	55.14	0.65 $\pm$ 0.01	0.39	0.80 $\pm$ 0.00	0.62	0.799 $\pm$ 0.01	0.62

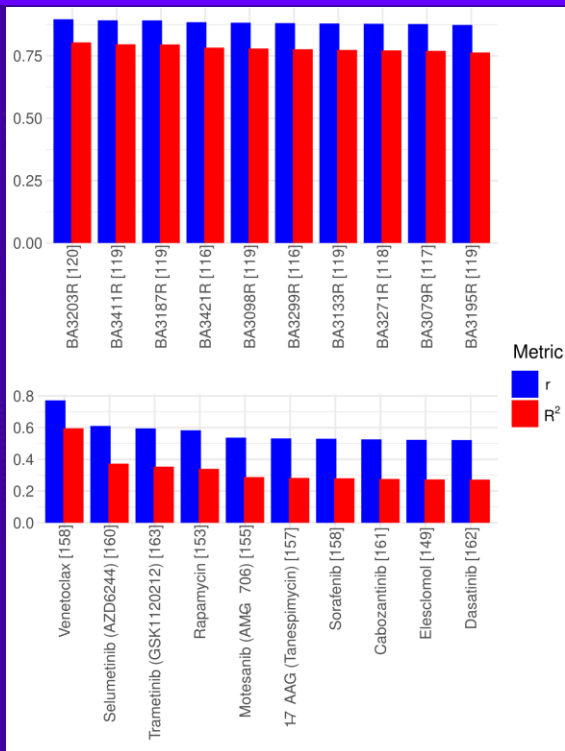
Table 1. Performance comparison of ML models predicting drug response for AML patients. Here we highlight the model with the best prediction metrics on the test set.

PT-AML: Results

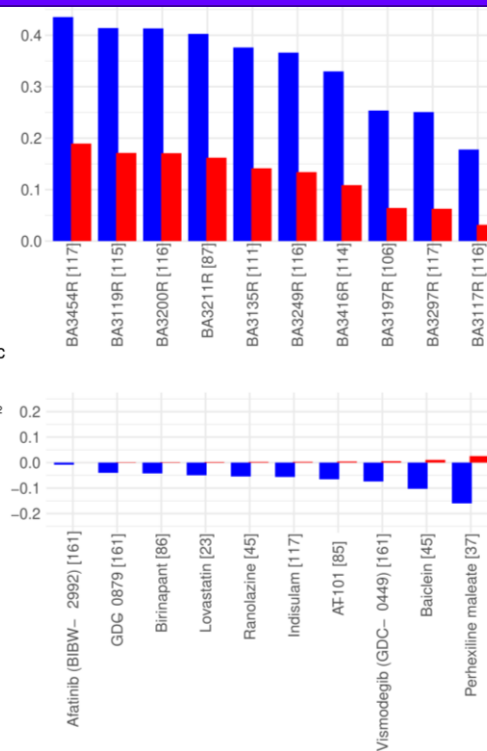
Top/Worst Profiles



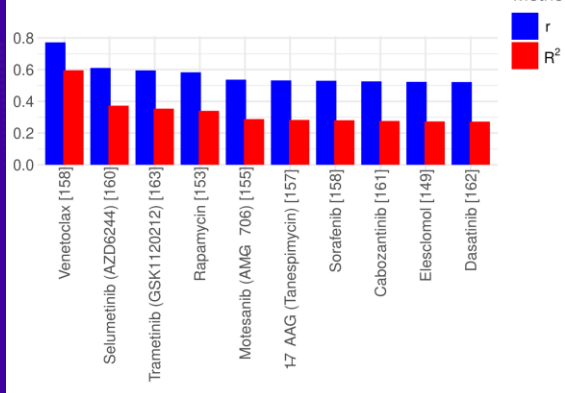
Top Best Patient Predictions



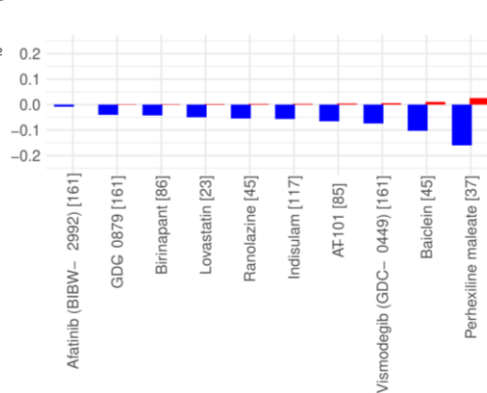
Worst Patient Predictions



Top Best Drug Predictions



Worst Drug Predictions



- PT-AML: Machine Learning framework for personalized drug responses in AML
- Leverages BeatAML cohort for drug & multi-omics patient representations
- CatBoost-based model shows strong performance on independent test set
- Potential to guide therapy selection, pending further clinical validation

Future

It's already here

TxT Prompt

Instructions: Answer the following question about **drug synergy**

Context: Synergy is a dimensionless measure of deviation of an observed drug combination response from the expected effect of non-interaction. Drug combinations were tested against cancer cell lines...

Question: Given two **drug SMILES strings** and a **cell line description**, predict the normalized **drug synergy** from 000 to 1000, where 000 is minimum **drug synergy** and 1000 is maximum **drug synergy**.

Drug1 SMILES: **C(=O)(N)NO**

Drug2 SMILES: **N.N.Cl[Pt+2]Cl**

Cell line description: **786-O**, **renal cell adenocarcinoma**

Answer: 562

Drug1 SMILES: **CS(=O)(=O)CCCCO**

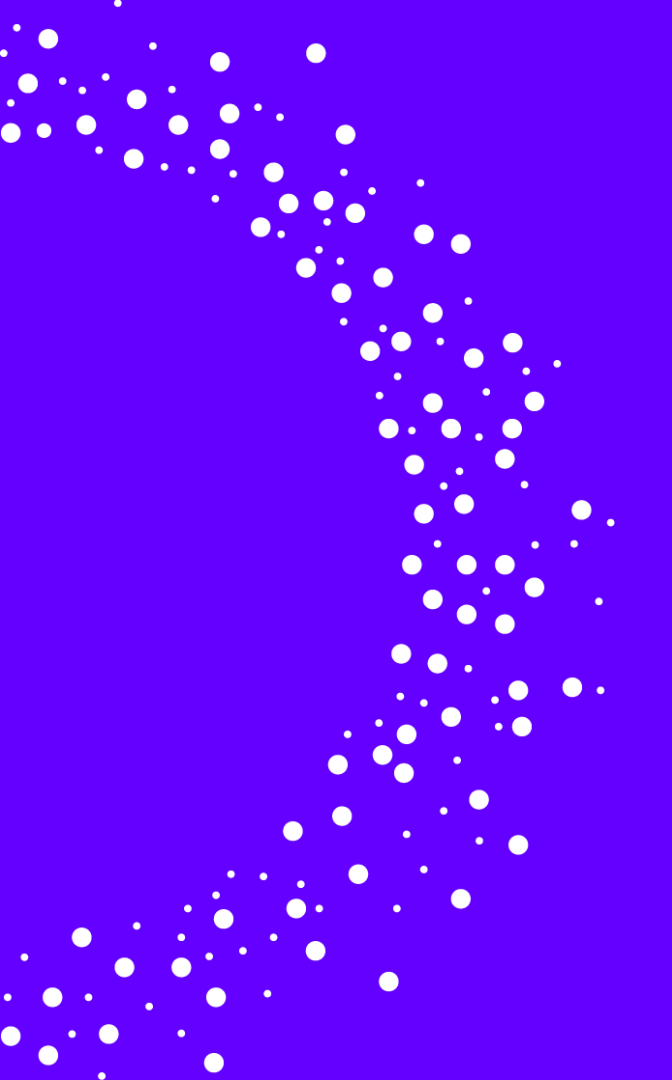
Drug2 SMILES: **N.N.Cl[Pt+2]Cl**

Cell line description: **OVCAR3**, **ovarian cell adenocarcinoma**

Answer: 678

...

- ChatGPT like model takes cell line and drugs as input and outputs drug response
- Tx-LLM developed by Google DeepMind



Q & A